The Business Case for Artificial Intelligence

How AI Can Improve Efficiency, Decision-Making, and Organizational Performance

A B2B Technology White Paper

Abstract

This white paper presents a business case for artificial intelligence grounded in measurable outcomes rather than hype. It argues that AI creates the greatest organizational value when it augments human expertise rather than replacing it: automating appropriate repetitive work, improving data analysis and decision support, strengthening knowledge management, and extending customer and employee support capacity. Drawing on peer-reviewed research, government frameworks, and independent industry studies, the paper examines realistic applications in documentation, accessibility, and operational scaling, and outlines methods for measuring return on investment. It also addresses the risks of poorly governed AI adoption, including factual errors, data privacy exposure, and cybersecurity threats, and proposes a practical, pilot-based framework for responsible implementation. The central conclusion is that organizations create more durable value by asking how AI can help their people work better, rather than how many people AI can replace.

Executive Summary

Artificial intelligence has moved from experimentation to mainstream business use faster than most prior enterprise technologies. Generative AI in particular reached an estimated 53 percent of the global population within roughly three years of ChatGPT's release, a faster adoption curve than either the personal computer or the internet (Stanford HAI, 2026). Yet rapid adoption has not translated evenly into business value. An MIT Media Lab study of more than 300 enterprise AI deployments found that 95 percent of generative AI pilots produced no measurable effect on profit and loss, while a small minority produced millions of dollars in value (MIT NANDA, 2025). The difference was rarely the underlying model. It was whether the organization tied the technology to a specific, well-defined business problem, integrated it into an existing workflow, and measured the result.

This paper makes the case that AI delivers its greatest business value when it augments human expertise rather than replacing it. Independent, peer-reviewed research supports this framing. A randomized controlled trial of professional writing tasks found that access to a generative AI assistant reduced completion time by roughly 40 percent while raising quality scores by about 18 percent, with the largest gains among lower-performing writers (Noy & Zhang, 2023). A separate study of more than 5,000 customer support agents at a Fortune 500 company found a 14 to 15 percent increase in issues resolved per hour, again concentrated among newer and less experienced staff (Brynjolfsson, Li, & Raymond, 2023, 2025). In both cases, AI functioned as a performance equalizer that raised the floor for less experienced workers rather than a replacement for expert judgment.

The paper reviews specific, realistic applications across automation of repetitive work, decision support, knowledge management, customer and employee support, technical documentation, and accessibility. It then addresses how organizations can establish baseline metrics and measure ROI, and provides a substantive discussion of risk: factual errors and hallucination, data privacy, cybersecurity, intellectual property, bias, and regulatory compliance, referencing the NIST AI Risk Management Framework and the OECD AI Principles. It closes with a practical, incremental framework for responsible AI adoption. Throughout, the paper avoids the assumption that AI's primary purpose is workforce reduction, and instead treats responsible governance as part of how ROI is achieved rather than an obstacle to it.

Introduction: Moving Beyond the AI Hype

Business coverage of artificial intelligence tends toward two extremes: claims that AI will transform every function overnight, and warnings that it exists primarily to eliminate jobs. Neither framing is useful for a decision-maker evaluating whether and how to invest. A more productive starting point is to treat AI the way organizations treat any other capital investment: by identifying a specific business problem, estimating the value of solving it, and testing whether a given tool actually solves it under real conditions.

The evidence on generative AI's economic potential is substantial but uneven in how it plays out inside individual organizations. McKinsey's analysis of 63 generative AI use cases across 16 business functions estimated an annual economic impact in the range of 2.6 to 4.4 trillion dollars globally, concentrated in functions such as customer operations, marketing and sales, software engineering, and research and development (McKinsey Global Institute, 2023). At the same time, McKinsey's 2024 survey work found that while 65 percent of organizations reported regularly using generative AI in at least one business function, most were still realizing value in only a narrow set of use cases (McKinsey, 2024). The gap between aggregate economic potential and realized value at the organizational level is the central problem this paper addresses. The MIT NANDA findings referenced above make the same point from a different angle: broad adoption of generic AI tools is common, but converting that adoption into measurable financial return is not, and the organizations that succeed tend to evaluate AI against specific workflows rather than adopting it as a general-purpose initiative (MIT NANDA, 2025).

This paper's premise is that the business case for AI should be built use case by use case, with clearly defined success metrics established before implementation, and with realistic expectations about what current AI systems can and cannot do reliably.

Automating Repetitive Work

AI is most reliable, and least controversial, when applied to well-defined, repetitive cognitive and administrative tasks: drafting routine correspondence, summarizing meeting notes, extracting data from structured documents, categorizing support tickets, or generating first-draft reports from templated inputs. These tasks share two characteristics that make them good candidates for AI assistance: they are high in volume and low in ambiguity, and the cost of an occasional error is manageable when a human reviews the output before it is finalized.

The productivity effects of this kind of assistance have been measured directly. In the randomized trial described above, mid-level professionals writing occupation-specific documents such as press releases, grant applications, and short reports completed the tasks roughly 40 percent faster with generative AI assistance, and independent reviewers rated the AI-assisted output about 18 percent higher in quality (Noy & Zhang, 2023). A separate line of research on software development found that developers using an AI coding assistant completed a defined programming task in roughly half the time of a control group (Peng et al., 2023, as cited in the arXiv survey by an interdisciplinary research team, 2023). These are controlled studies of specific, bounded tasks; they are not evidence that AI can perform an entire job function without oversight. The realistic business application is narrower and more durable: use AI to produce a first draft or a structured extraction, and route the result to the employee who previously did the work from scratch, freeing their time for review, judgment calls, and exceptions that fall outside the routine pattern.

Improving Data Analysis and Decision Support

A second category of value involves helping employees process more information than they could review manually and surface patterns that inform a decision. This is meaningfully different from letting AI make the decision itself. In a decision-support application, an AI system might summarize customer feedback across thousands of records, flag anomalies in transaction data, or generate a first-pass comparison of vendor proposals against defined criteria. A human decision-maker then reviews that output, weighs it against context the system does not have, and makes the call.

The distinction matters because the reliability of AI-generated analysis varies by task and cannot be assumed to be uniformly high. Stanford's 2026 AI Index Report, an independent annual review produced by Stanford's Institute for Human-Centered AI, found that hallucination rates on a new accuracy benchmark ranged from 22 to 94 percent across 26 leading models, and that model accuracy dropped sharply when a false premise was framed as something the user personally believed rather than something a third party claimed (Stanford HAI, 2026). This is precisely why decision support, not autonomous decision-making, is the appropriate framing for most business applications today: the system can accelerate the gathering and organizing of information, but a person with domain knowledge should verify material conclusions before they inform a business decision, particularly one with financial, legal, or safety consequences.

Improving Organizational Knowledge Management

Large organizations routinely lose time to a problem that predates generative AI: institutional knowledge scattered across systems, documents, and individual employees' memory, with no reliable way to search across all of it at once. McKinsey Global Institute's earlier research on knowledge work estimated that employees spent roughly a fifth of their working time, the equivalent of one day per week, searching for and gathering information needed to do their jobs (McKinsey Global Institute, 2023, citing 2012 research).

AI-powered enterprise search and retrieval tools address this problem directly by indexing internal documentation, prior project records, support tickets, and policy documents, and returning natural-language answers grounded in that internal content rather than the open internet. Applied well, this reduces the time employees spend locating existing answers and reduces duplicated work when the same question has already been answered elsewhere in the organization. It also creates a partial safeguard against knowledge loss when an experienced employee leaves, though it does not replace the judgment that employee applied when the underlying information was ambiguous or incomplete. The reliability of this application depends heavily on the quality and currency of the underlying document set; a retrieval system built on outdated or contradictory source material will surface outdated or contradictory answers with equal confidence.

Improving Customer and Employee Support

Customer and employee support functions offer some of the most rigorously studied applications of generative AI to date. In the Brynjolfsson, Li, and Raymond study of more than 5,000 customer support agents at a Fortune 500 software company, agents given access to a generative AI conversational assistant resolved 14 to 15 percent more issues per hour on average, with gains concentrated among newer and lower-skilled agents, whose productivity improved by roughly a third; highly experienced agents saw comparatively small effects (Brynjolfsson, Li, & Raymond, 2023, published in the Quarterly Journal of Economics, 2025). The researchers also found that agents followed the AI's suggested responses only about 38 percent of the time, evidence that the tool functioned as a recommendation the agent could accept or override rather than an autonomous responder, and that customer sentiment and agent retention both improved.

The appropriate design for this kind of deployment reflects that evidence: an AI assistant surfaces a recommended response or relevant documentation, a human agent decides whether to use it, and clearly defined criteria route complex, sensitive, or emotionally charged interactions to a human without AI mediation. The same principle applies to internal employee-facing support, such as IT help desks or HR inquiries, where AI can resolve high-volume, well-defined requests (password resets, benefits enrollment questions, policy lookups) while escalating ambiguous or high-stakes cases to a specialist.

AI and Documentation

Technical writing and documentation teams have particular reason to evaluate AI carefully, both because documentation work involves many of the repetitive tasks AI handles well and because documentation errors carry direct downstream cost. AI can reasonably assist with content classification and tagging, summarizing long source material into shorter reference documents, generating first-draft explanations of a process from structured inputs such as API specifications or engineering change orders, identifying where similar content already exists to support reuse, maintaining consistent terminology across a document set, flagging documentation that has not been updated to reflect a recent product or process change, and highlighting apparent gaps where a described process lacks a corresponding procedure.

None of these applications remove the documentation team's core responsibility. Technical accuracy, validation against the actual system or process being documented, usability testing with real users, and final editorial sign-off remain human responsibilities that AI does not perform reliably on its own. This is not a hedge; it follows directly from the accuracy limitations described in the previous sections. A documentation team that uses AI to accelerate drafting and search while retaining full ownership of verification and final review captures the efficiency benefit without inheriting the accuracy risk.

Accessibility and AI

AI-based tools can support several legitimate accessibility applications: automated transcription and captioning of audio and video content, translation of documentation and support material into additional languages, adaptation of dense technical text into plainer language, and multimodal interfaces that let users interact with information through voice or alternative input methods. Each of these can materially expand who is able to access an organization's content and services.

These tools also have known limitations that determine whether they genuinely improve accessibility or merely appear to. Automated captions and transcriptions can misrepresent specialized terminology, accents, or overlapping speech; automated translation can alter technical meaning in ways a non-native reader has no way to detect; and plain-language rewrites can inadvertently drop information that a user with a disability specifically needed. Because of this, AI-generated accessibility output benefits from the same review discipline as any other AI-generated content: it should be treated as a draft that a qualified reviewer, ideally one following recognized accessibility standards, checks before it is treated as the accessible version of the content. AI should be understood as a tool that can reduce the cost and time of producing accessible content, not as a technology that automatically makes content accessible on its own.

Scalability and Operational Efficiency

Organizations often frame AI-driven efficiency purely in terms of headcount reduction. A more durable framing treats AI as a way to increase the capacity of an existing team to handle a growing volume of work, or to shift the composition of that team's time toward higher-value activity, without necessarily reducing the team's size. In a support organization, for example, AI-assisted triage and drafting can allow the same number of agents to handle a higher case volume during a demand spike, or can allow experienced staff to spend more time on complex escalations and mentoring newer employees rather than routine ticket volume, an application consistent with the skill-leveling effect Brynjolfsson, Li, and Raymond documented in customer support (2025).

This distinction between capacity expansion and headcount reduction is not merely a matter of framing; it reflects where the research shows AI adds the most value. The productivity gains documented across the studies cited in this paper are concentrated in tasks AI performs reliably, freeing human time for tasks that still require judgment, relationship management, and contextual decision-making that current AI systems cannot reliably replicate. Workflow redesign, not headcount arithmetic, is where most of the realized value described in this paper actually originates.

Measuring Business Value and ROI

The gap between AI's aggregate economic potential and its inconsistent realized value inside individual organizations, as documented by MIT NANDA (2025), points to a specific and correctable cause: many AI initiatives are launched without a defined success metric or a pre-implementation baseline against which to measure it. Establishing that baseline before deployment is what allows an organization to say, with evidence, whether a given AI investment produced value.

Metrics appropriate to an AI initiative should be chosen based on the specific business problem the initiative targets, and typically include some combination of the following: time required to complete a defined task or process; cost per transaction or per resolved case; average resolution time for support or service requests; time required to create or update a piece of documentation; broader employee productivity indicators tied to the workflow in question; customer satisfaction scores; error or defect rates before and after deployment; time required to search for or retrieve internal information; and, where the use case is customer-facing, conversion rate or revenue impact. Baseline measurement should occur before the AI tool is introduced, using the same definitions and data sources that will be used to measure the result afterward, so the comparison is valid.

MIT NANDA's research on why most pilots fail to show a financial return offers a useful diagnostic for organizations building their own measurement approach: the successful minority of organizations in that study tended to buy or build tools that integrated into a specific, already-defined workflow and adapted to organizational context over time, while the unsuccessful majority deployed general-purpose tools that were easy to demonstrate but never became load-bearing parts of an actual process (MIT NANDA, 2025). A measurement plan built around a specific workflow, rather than around a technology category, is more likely to produce a defensible ROI figure.

Risks and Responsible Implementation

Responsible implementation is not a constraint on realizing AI's business value; based on the evidence reviewed in this paper, it is a precondition for it. The following risk categories warrant explicit attention in any AI adoption plan.

Hallucinations and Factual Errors

Generative AI systems can produce fluent, confident output that is factually incorrect, a behavior commonly called hallucination. Stanford HAI's 2026 AI Index Report found hallucination rates ranging from 22 to 94 percent across 26 leading models on a benchmark designed to test whether a model could distinguish an established fact from a false claim the user presented as their own belief, and noted that documented AI-related incidents rose from 233 in 2024 to 362 in 2025 (Stanford HAI, 2026). The evidence across studies is not fully consistent on a single hallucination rate, because rates vary substantially by task, domain, and how a model is grounded in source material; what the research consistently shows is that the risk is real, task-dependent, and not eliminated by model quality alone. This is the central justification for human review of AI output before it is used in a business decision, published externally, or relied upon for compliance purposes.

Data Privacy and Cybersecurity

IBM's 2025 Cost of a Data Breach Report, based on research conducted with the Ponemon Institute across more than 600 breached organizations, found that breaches involving unsanctioned “shadow AI” tools cost an average of 670,000 dollars more than breaches without that factor, and that 63 percent of breached organizations had no AI governance policy in place at the time of the incident (IBM Security, 2025). The same report found that organizations with mature, well-governed AI-enabled security operations resolved incidents meaningfully faster and at lower cost, indicating that the risk originates specifically from ungoverned AI use rather than from AI use in general. Organizations should have clear policy on which AI tools are approved for use, what categories of data may be submitted to them, and how those tools are monitored.

Intellectual Property

Questions about the copyright status of AI training data and AI-generated output remain legally unsettled in multiple jurisdictions. Organizations should treat AI-generated content intended for external publication, and any AI tool that will process proprietary or client-confidential material, as requiring legal review of applicable terms of service and ownership provisions before broad deployment.

Bias and Fairness

AI systems trained on historical data can reproduce or amplify patterns of bias present in that data, particularly in applications that touch hiring, lending, performance evaluation, or other decisions affecting individuals. This risk is a central concern of both the NIST and OECD frameworks discussed below and warrants specific testing before an AI system is used in any decision with material consequences for individuals.

Governance, Transparency, and Regulatory Compliance

Two widely referenced frameworks provide useful structure for AI governance. The NIST AI Risk Management Framework, released by the U.S. National Institute of Standards and Technology in January 2023, organizes risk management around four functions, Govern, Map, Measure, and Manage, and defines trustworthy AI in terms of validity and reliability, safety, security and resilience, accountability and transparency, explainability, privacy, and fairness (NIST, 2023). It is voluntary and designed to be adapted through organization-specific profiles rather than applied as a fixed checklist. The OECD AI Principles, first adopted in 2019 and updated in May 2024 to address generative AI specifically, are endorsed by 47 governments and organize responsible AI development around five value-based principles: inclusive growth and well-being, respect for human rights and democratic values, transparency and explainability, robustness and safety, and accountability (OECD, 2024). Organizations operating across jurisdictions should also track sector-specific and regional regulation, such as the EU AI Act's risk-based obligations, which continues to evolve.

Human Oversight and Employee Training

Every risk category above points to the same operational requirement: defined points of human review built into the workflow before AI output reaches a customer, a regulator, or a business decision, and employees trained to recognize where AI output is more or less likely to be reliable. This oversight function is not a one-time setup task; it should be revisited as AI tools, source data, and business processes change.

A Practical Framework for AI Adoption

The following incremental sequence reflects both the governance principles above and the evidence, drawn from MIT NANDA's study of successful and unsuccessful pilots, that the deciding factor in AI value creation is close integration with a specific workflow rather than broad, general-purpose deployment (MIT NANDA, 2025).

- Identify a specific business problem with a clear owner and a defined workflow, rather than starting from a technology category.
- Establish baseline metrics for that workflow before any AI tool is introduced, using data sources that will remain available for post-implementation comparison.
- Determine whether AI is an appropriate solution to the identified problem, including whether the task tolerates the error rates documented in this paper and whether adequate source data exists.
- Conduct a limited, time-boxed pilot with a defined group of users and a defined success threshold agreed upon in advance.
- Establish human oversight appropriate to the risk level of the task, with explicit review points before AI output reaches a customer, a decision-maker, or a compliance-relevant record.
- Measure results against the pre-established baseline and success threshold, using the same metric definitions established at the outset.

- Identify problems surfaced during the pilot, including accuracy issues, workflow friction, and user resistance, and address them directly rather than treating early friction as a reason to abandon the initiative prematurely.
- Improve the workflow based on pilot findings, which may include retraining users, adjusting the tool's scope, or improving the underlying source data.
- Scale only when the pilot has demonstrated measurable value against the baseline; treat pilots that have not met their threshold as informative rather than as failures to be quietly repeated without changes.

Conclusion: Human Expertise Plus AI Capability

The research reviewed in this paper points in a consistent direction. AI delivers measurable, reproducible value when it is applied to specific, well-understood tasks, integrated into an existing workflow, paired with human review appropriate to the risk involved, and measured against a baseline established before implementation. It delivers considerably less value, and carries meaningfully more risk, when it is adopted broadly on the assumption that its purpose is to reduce headcount, without a defined problem, without governance, and without a way to measure whether it worked.

Organizations planning their AI strategy are better served by asking, for each candidate use case, how AI can help their people perform their work better, rather than asking how many people AI can replace. The evidence reviewed here, from randomized trials of professional writing and customer support to large-scale studies of enterprise AI deployment and governance, supports the former question as the one more likely to produce a durable, measurable business return.

Key Takeaways

- AI creates the most reliable business value when applied to specific, well-defined tasks with an established baseline, not as a broad, undifferentiated initiative.
- Controlled studies show meaningful productivity gains from AI assistance in professional writing (about 40 percent faster, 18 percent higher quality) and customer support (14 to 15 percent more issues resolved per hour), concentrated among less experienced workers.
- An MIT study of more than 300 enterprise AI deployments found 95 percent of pilots produced no measurable P&L impact, largely because tools were not integrated into specific workflows or measured against a baseline.
- AI's factual reliability varies substantially by task and is not uniformly high; independent benchmarking in 2026 found hallucination rates ranging from 22 to 94 percent depending on the model and test condition, underscoring the need for human review.
- Ungoverned AI use carries measurable cybersecurity cost: breaches involving shadow AI cost an average of 670,000 dollars more, and most breached organizations lacked an AI governance policy.
- The NIST AI Risk Management Framework and the OECD AI Principles offer established, non-proprietary structures for AI governance covering transparency, accountability, safety, and fairness.

- A practical, staged framework, defined problem, baseline metrics, limited pilot, human oversight, measured results, then scale, produces more defensible outcomes than broad, unmeasured deployment.

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